

Memory MAC Channels — Results

Chained NPD for Two-User MAC with Memory

April 2026

1 ISI-MAC

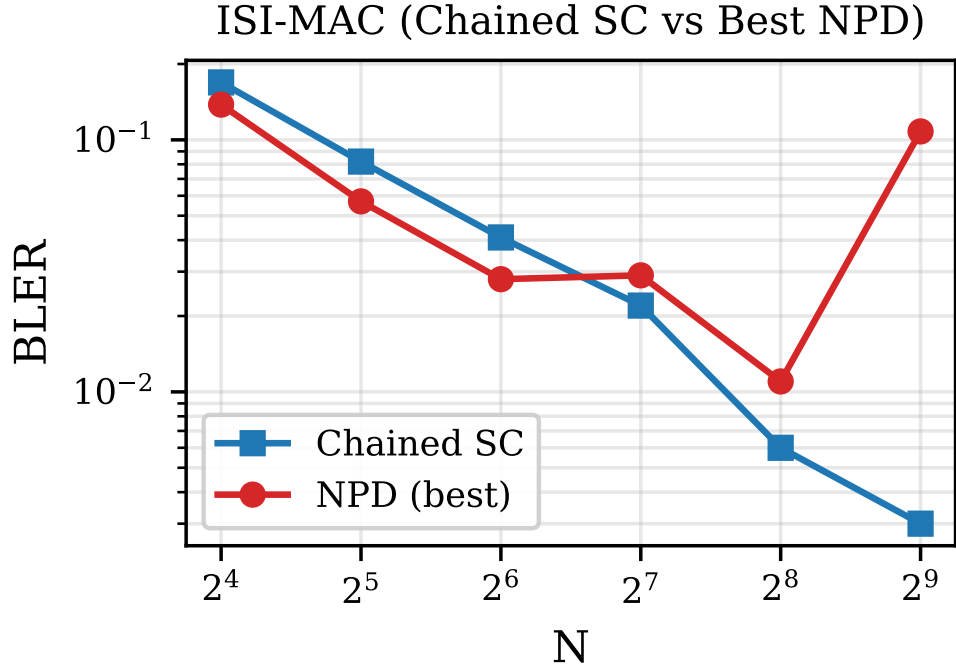
$Z_i = (1-2X_i) + (1-2Y_i) + 0.3((1-2X_{i-1}) + (1-2Y_{i-1})) + W_i$, SNR = 6 dB, $\sigma^2 = 0.251$.

Path: `make_path(N, N)` (Class C corner). State space $|S| = 4$.

Decoder: chained NPD (Stage 1: U , Stage 2: $V|\hat{U}$). BiGRU `z.encoder`.

Baseline: chained trellis SC (2-stage, $|S| = 2$ per stage).

N	k_u/k_v	Chained SC	NPD (best)	Architecture	NPD/SC
16	4 / 7	0.169	0.138	d=16 h=100	0.82 ×
32	7 / 15	0.082	0.057	d=16 h=100	0.69 ×
64	15 / 29	0.041	0.028	d=16 h=100	0.68 ×
128	30 / 58	0.022	0.029	d=64 h=128	1.32×
256	59 / 117	0.006	0.011	d=64 h=128	1.83×
512	119/233	0.003	0.108	d=64 h=128	43 ×



2 MA-AGN MAC

$Z_i = (1-2X_i) + (1-2Y_i) + N_i$, $N_i = \alpha \cdot N_{i-1} + W_i$ (AR(1) noise).

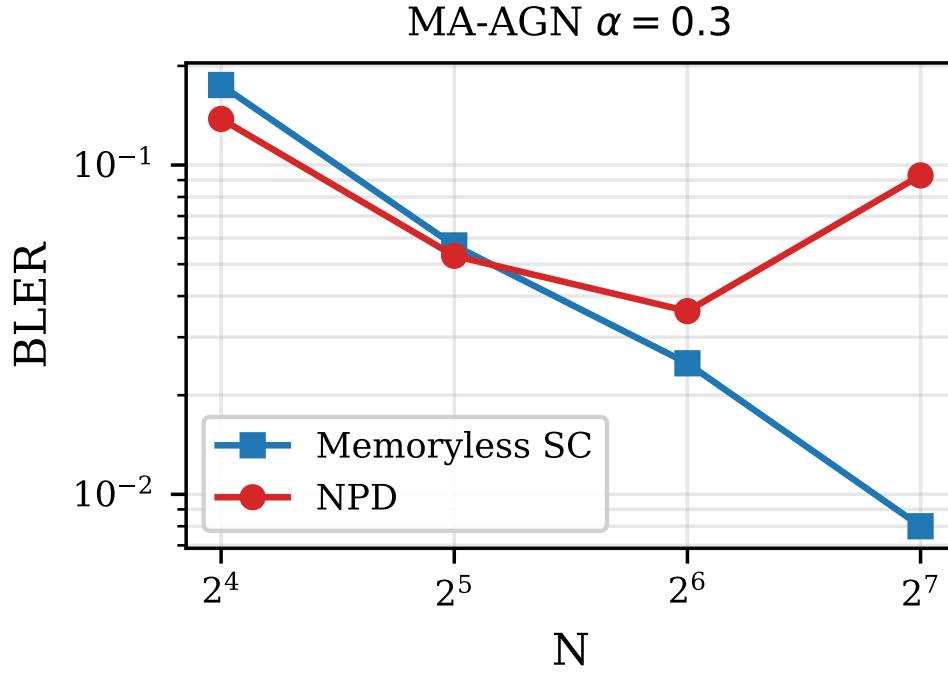
SNR = 6 dB. Continuous state (no analytical trellis).

Path: `make_path(N, N)` (Class C corner).

Baseline: memoryless GMAC SC (ignores noise correlation).

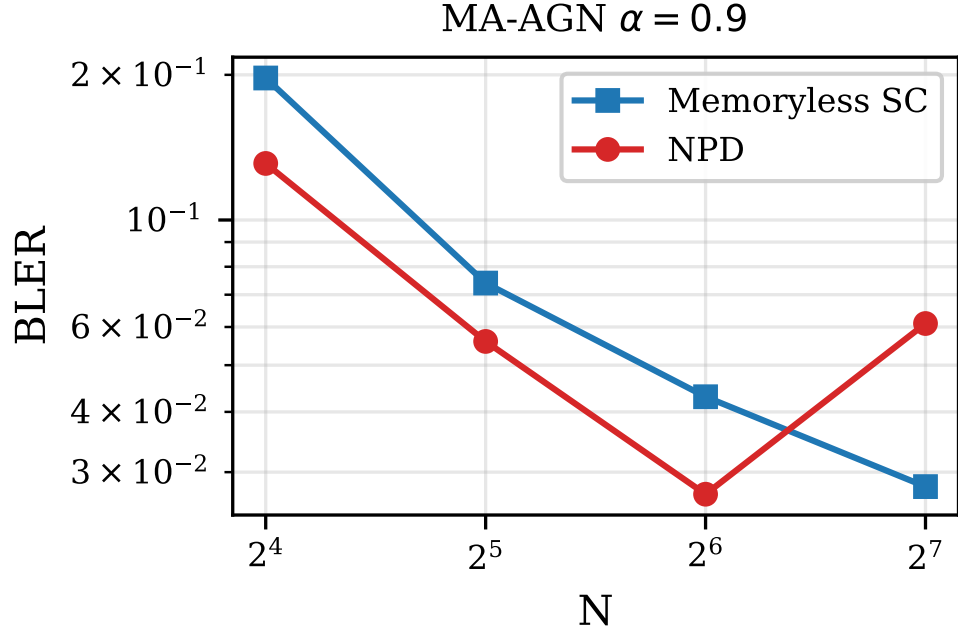
$\alpha = 0.3$ (weak memory)

N	k_u/k_v	Memoryless SC	NPD (d=16 h=100)	NPD/SC
16	4 / 7	0.175	0.138	0.79×
32	7 / 15	0.057	0.053	0.93×
64	15 / 29	0.025	0.036	1.42×
128	30 / 58	0.008	0.093	11.6×



$\alpha = 0.9$ (strong memory)

N	k_u/k_v	Memoryless SC	NPD (d=16 h=100)	NPD/SC
16	4 / 7	0.197	0.131	0.66×
32	7 / 15	0.074	0.056	0.76×
64	15 / 29	0.043	0.027	0.62×
128	30 / 58	0.028	0.061	2.18×



Summary

Channel	NPD vs baseline	N range
ISI-MAC (d=16 h=100)	0.68–0.82 $\times$ chained trellis SC	$N = 16$ –64
ISI-MAC (d=64 h=128)	1.3–1.8 $\times$ chained trellis SC	$N = 128$ –256
MA-AGN $\alpha = 0.9$	0.62–0.76 $\times$ memoryless SC	$N = 16$ –64
MA-AGN $\alpha = 0.3$	0.79 $\times$ memoryless SC	$N = 16$ only